

# SENSEI: A Simple, Elegant, and Nimble Subtitle Editing Interface

Mauro Cettolo<sup>1†</sup>, Elia Soliman<sup>2†</sup>, Roldano Cattoni<sup>1</sup>, Matteo Negri<sup>1</sup>, Luisa Bentivogli<sup>1</sup> and Alessio Palmero Aprosio<sup>2\*</sup>  
<sup>1</sup> Fondazione Bruno Kessler, Trento · <sup>2</sup> University of Trento — <sup>†</sup> Equal contribution \* Corresponding author

## 1 MOTIVATION

Automatic subtitling keeps improving, but broadcast-ready output still needs human validation against reading speed, line length, lines-per-block, and text accuracy. Existing editors are either generic and free, or polished and closed. SENSEI closes that gap.

## 2 KEY CONTRIBUTIONS

- **Open by design.** Apache-2.0, self-hostable, web-based — no local install, usable from any browser.
- **Natively bilingual.** source and target are first-class, reactively synchronized tracks.
- **Compliance built in.** live, per-block, user-configurable monitoring of every formatting constraint.

## 3 SYSTEM ARCHITECTURE

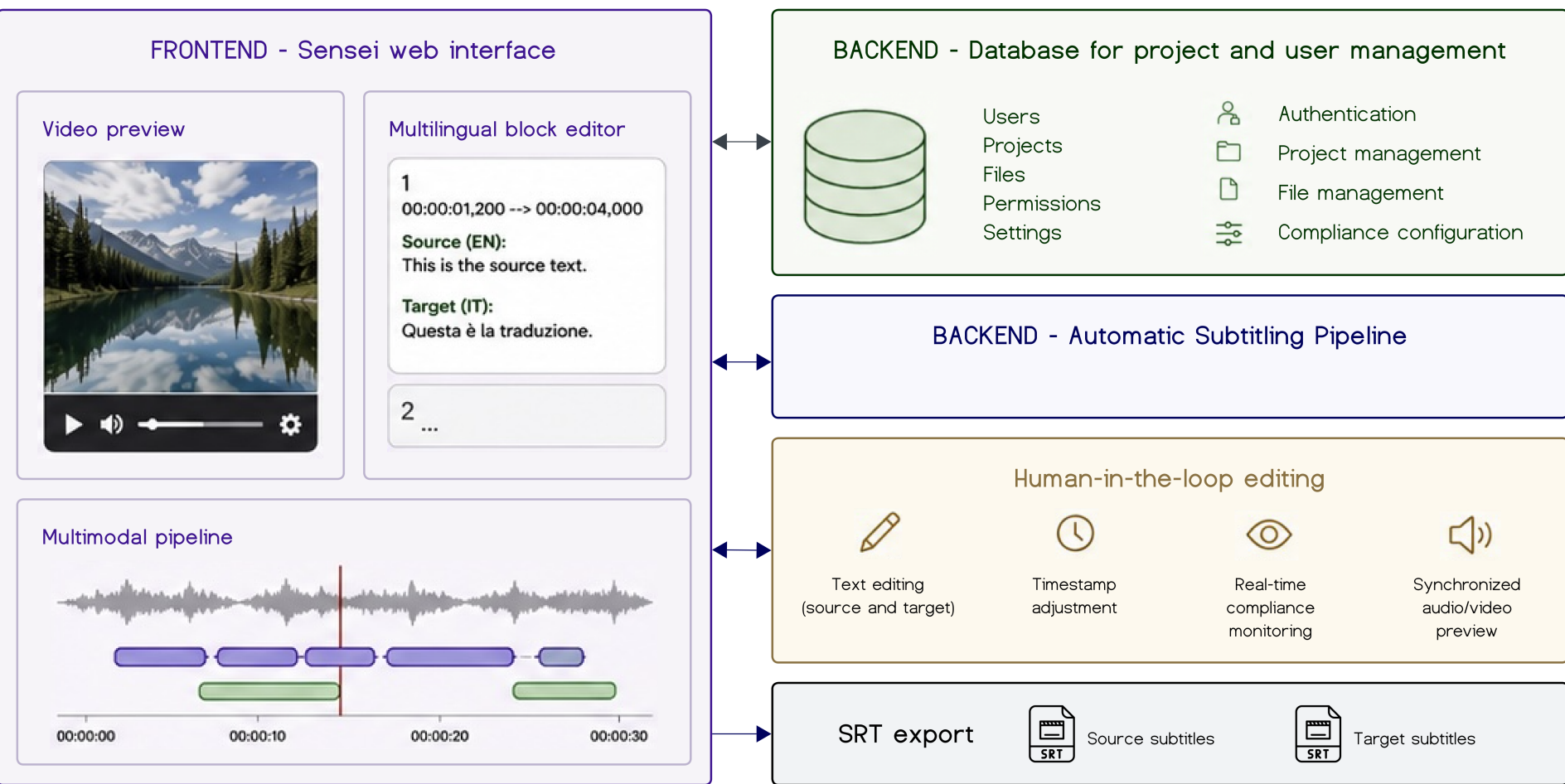
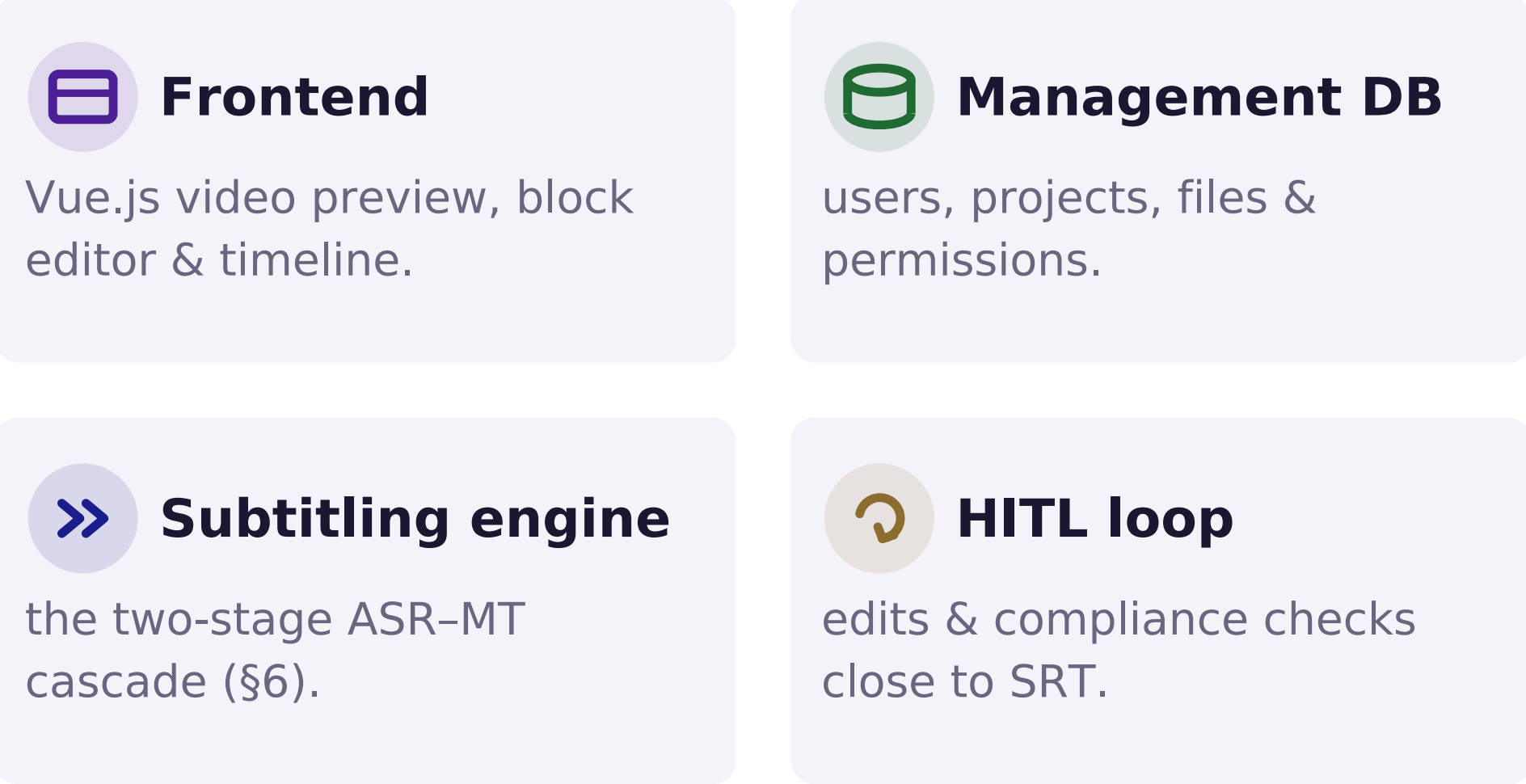


Figure 1. Overall architecture of the SENSEI ecosystem.



## 8 HOW DOES SENSEI COMPARE?

The only environment combining automatic generation, bilingual tracks and compliance-aware editing under a fully open license:

| Functions                     | SENSEI | MateSub                 | HappyScribe           | Aegisub | Subtitle Edit |
|-------------------------------|--------|-------------------------|-----------------------|---------|---------------|
| Automatic subtitle generation | ✓      | ✓                       | ✓                     | □       | ✓             |
| Merge                         | ✓      | ✓                       | ✓                     | ✓       | ✓             |
| Add                           | ✓      | ✓                       | ✓                     | ✓       | ✓             |
| Delete                        | ✓      | ✓                       | ✓                     | ✓       | ✓             |
| Duplicate                     | ✓      | □                       | □                     | ✓       | □             |
| Edit text                     | ✓      | ✓                       | ✓                     | ✓       | ✓             |
| Edit timestamp (keyboard)     | ✓      | ✓                       | ✓                     | ✓       | ✓             |
| Edit timestamp (mouse)        | ✓      | ✓                       | ✓                     | □       | ✓             |
| Fullscreen video              | ✓      | □                       | ✓                     | □       | ✓             |
| Resizable interface           | ✓      | open/close sidebar only | sidebar only          | ✓       | ✓             |
| Multi-language                | ✓      | □                       | separate environments | □       | □             |
| Load SRT file                 | ✓      | ✓                       | ✓                     | ✓       | ✓             |

### Funding

This work has received funding from the European Union's Horizon Europe programme under grant agreement No. 101213369 (DVPS).

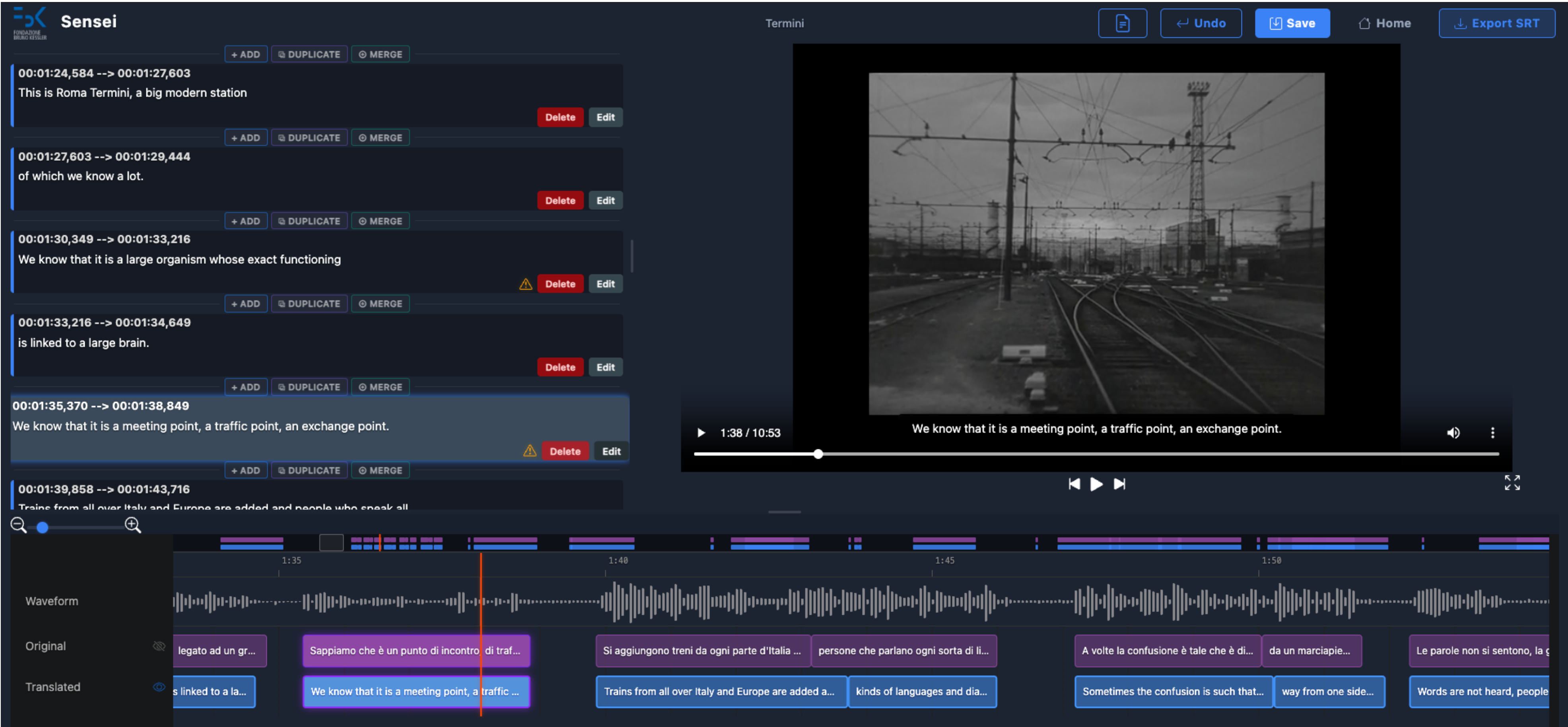
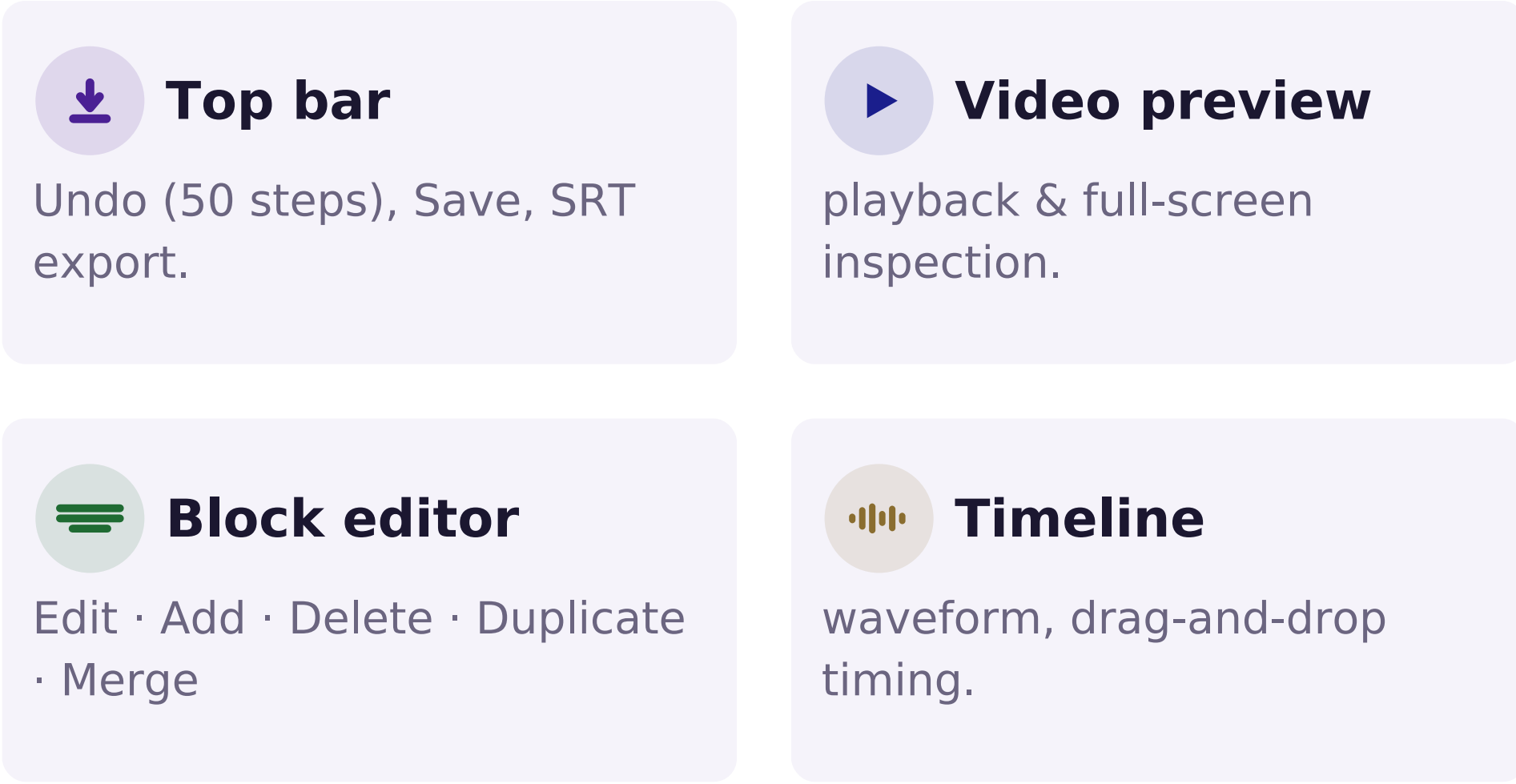
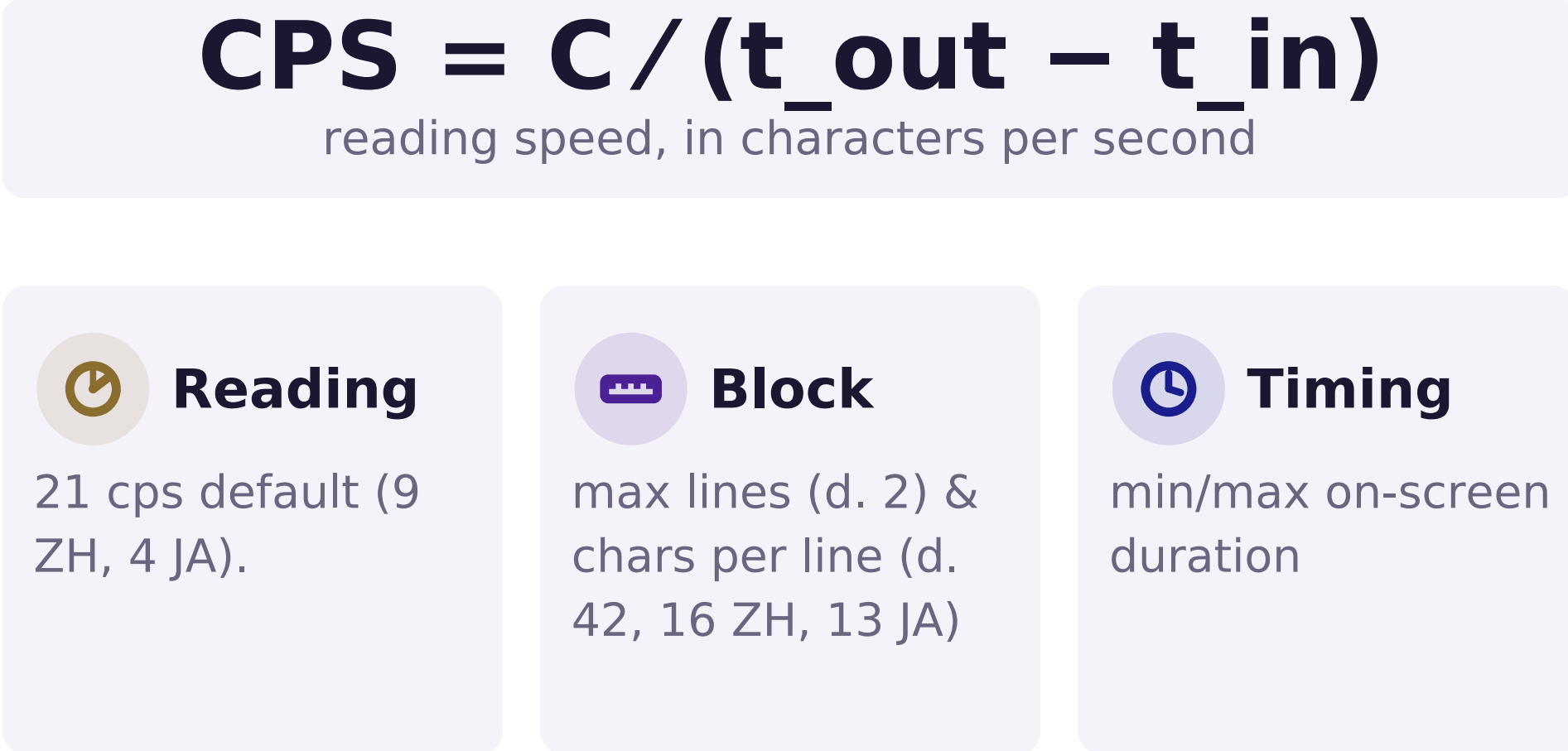


Figure 3. The SENSEI workspace, fully synchronized live.

## 4 THE EDITING INTERFACE



## 5 COMPLIANCE AT A GLANCE



All thresholds are user-editable; violations are flagged live.

## 6 SUBTITLING PIPELINE

A two-stage ASR-MT cascade, built entirely from freely available components.

### STAGE 1 — BASELINE



### STAGE 2 — SENTENCE-AWARE REFINEMENT



## 7 EVALUATION HIGHLIGHTS

- IWSLT 2026 Subtitling task, 3 domains, up to 5 languages: the sentence-aware cascade consistently beats the baseline on BLEU, ChrF and BLEURT.
- On ITV test2023: competitive with AppTek's commercial system, ahead of all previous FBK submissions (SubER).